

Header
Data point groups
Data points
Parameter Description
Custodian
Dig-site
Baseline Mandatory / Baseline Non-Mandatory / Enhanced
Baseline / Enhanced?
Input Format
Allowed Values / Code List
Quality Rules / Validation
Granularity (Grain)
Time Dimension
Example
Yield Optimizer Produce Driver
Yield Optimizer Satellite Monitoring
Source System (SAP)
Source Table (SAP)

Data Point Groups
Grower Master Data
Farm Master Data
Contract
Operational
Financial
Field
Planting
Fertilizer
Chemicals
Irrigation
Crop development
Chipstock dig data

Storage Chemicals
Storage Infrastructure
Chipstock actual harvest
Chipstock Storage
Store cell check
Storage Fry Sample

Foundation Critical
Foundation Recommended
Advance Insights
Irrigation
Mandatory
Optional

Description
The major clusters under which all the data points are sub-classified into
The list of all parameters for which data needs to be collected
Description of the individual data points
Entity responsible for the data - whoever reports the data is the custodian
Sub-process of the Farm-to-fork flow under which the data point is captured
Classifying the data point as either Baseline or Enhanced; If baseline - sub-classify as Mandatory or Non-Mandatory
Classifying the data point as baseline (must have) and enhanced (good to have)
The format in which the input needs to be provided e.g. Year format, time format etc.
Acceptable list of values for the parameter
Input validations to be put in place to ensure user does not input an incorrect value
The atomicity at which data points values are collected
The frequency over which the data point is to be collected
A sample input for the data point
Is the data point a mandatory/optional field for YO Produce Optimizer or not applicable?
Is the data point a mandatory/optional field for YO Satellite Monitoring or not applicable?
The mapping in SAP tool to identify the tagging of the data point with the source system
The mapping in SAP tool to identify the tagging of the data point with the source table under the source system

Description
Core reference data describing a grower entity
Core reference data describing a farm entity
Data governing the commercial agreements
Non- agronomic category-such as batch no., loading date & time, planned delivery dates, plant, etc. and potato production data including input costs, contract values, yield, potato base amount, total amount, etc.
Non- agronomic category-such as batch no., loading date & time, planned delivery dates, plant, etc. and potato production data including input costs, contract values, yield, potato base amount, total amount, etc.
Individual Field data like ownership, type of irrigation, soil data, compaction data,
Planting dates, seed variety, seed lot details, seed rate, row spacing, field allocation
Fertilizer type- application method, timing, rates, and field-level records, irrigation system type, application volumes, timing, soil moisture readings and crop protection products.
Fertilizer type- application method, timing, rates, and field-level records, irrigation system type, application volumes, timing, soil moisture readings and crop protection products.
Fertilizer type- application method, timing, rates, and field-level records, irrigation system type, application volumes, timing, soil moisture readings and crop protection products.
Growth stages of the potato-emergence dates, % cover, tuber initiation etc.
Yield dig data like dig date, tuber weight, tuber count, % cover, harvested yield of potatoes total defects, measured solids, total solids etc.

Storage facilities data- including cell number, store name, fresh volume, stored volume, etc. anti-sprouting treatment details, Quality assessment data from fry testing, storage facility details like location & capacity, storage monitoring like rot, spoilage, conditions etc.
Storage facilities data- including cell number, store name, fresh volume, stored volume, etc. anti-sprouting treatment details, Quality assessment data from fry testing, storage facility details like location & capacity, storage monitoring like rot, spoilage, conditions etc.
Yield dig data like dig date, tuber weight, tuber count, % cover, harvested yield of potatoes total defects, measured solids, total solids etc.
Storage facilities data- including cell number, store name, fresh volume, stored volume, etc. anti-sprouting treatment details, Quality assessment data from fry testing, storage facility details like location & capacity, storage monitoring like rot, spoilage, conditions etc.
Storage facilities data- including cell number, store name, fresh volume, stored volume, etc. anti-sprouting treatment details, Quality assessment data from fry testing, storage facility details like location & capacity, storage monitoring like rot, spoilage, conditions etc.
Storage facilities data- including cell number, store name, fresh volume, stored volume, etc. anti-sprouting treatment details, Quality assessment data from fry testing, storage facility details like location & capacity, storage monitoring like rot, spoilage, conditions etc.

Baseline data template layer with mandatory core data elements required for Global KPI calculation
Baseline data template layer that supports better granularity, compliance, traceability and deeper analytics
Enhanced data template layer that supports advanced insights and innovation use cases
Parameters common with CFT
Mandatory parameter for Global KPI, Yield Optimizer Produce Driver or Satellite Monitoring
Optional parameter for Global KPI, Yield Optimizer Produce Driver or Satellite Monitoring

	51	51	51	51	51
Sl. No.	Data point groups	Data points	Parameter Description	Custodian	Dig-site
1	Farm Master Data	Field Name	Field ID or name of the farm	Agro	On-Farm Production
2	Farm Master Data	Variety	Potato variety name for Farm Master data	Agro	On-Farm Production
3	Grower Master Data	Country	ISO country code	Agro	On-Farm Production
4	Grower Master Data	Vendor Name	Name of the vendor/supplier	Agro	Commercial
5	Grower Master Data	Grower Name	Name of the grower/farmer	Agro	Commercial
6	Grower Master Data	Crop Season	Crop season year	Agro	On-Farm Production
7	Farm Master Data	Field Location	GPS/geographic field location via Polygon Mapping	Agro	On-Farm Production
8	Grower Master Data	Region	Name of the Geographic region or state	Agro	On-Farm Production
9	Grower Master Data	Vendor Contact details	Phone/Email of the vendor	Agro	Commercial

10	Storage Chemicals	CIPC/Chemical Applied	Sprout inhibitor application	Agro	Post Harvest Handling
11	Storage Chemicals	Other CIPC/Chemical Applied - Manual entry	Manual entry for other chemicals	Agro	Post Harvest Handling
12	Storage Chemicals	Application Method	How fertilizer was applied for storage chemicals	Agro	On-Farm Production
13	Chemicals	Product name	Pesticide/chemical product name	Agro	On-Farm Production
14	Chemicals	Product type	Type of plant protection product	Agro	On-Farm Production
15	Chemicals	Producer name	Chemical manufacturer name	Agro	On-Farm Production
16	Chemicals	Active ingredient	Active ingredient name	Agro	On-Farm Production
17	Chemicals	Liquid Chemical Application Rate	Application rate for liquids	Agro	On-Farm Production
18	Chemicals	Dry Chemical Application Rate	Application rate for dry products	Agro	On-Farm Production
19	Farm Master Data	PepsiCo Agronomist Name	Which PepsiCo Agronomist is collecting this data?	Agro	On-Farm Production

20	Farm Master Data	Irrigation type	Type of irrigation system	Agro	On-Farm Production
21	Fertilizer	N Applied	Nitrogen application rate	Agro	On-Farm Production
22	Fertilizer	N - Total	Total nitrogen applied	Agro	On-Farm Production
23	Fertilizer	P - Total	Total phosphorus applied	Agro	On-Farm Production
24	Fertilizer	K - Total	Total potassium applied	Agro	On-Farm Production
25	Fertilizer	Is digital Variable Rate Technology used?	VRT usage status	Agro	On-Farm Production
26	Fertilizer	Fertilizers are applied according to?	Basis for fertilizer application	Agro	On-Farm Production
27	Fertilizer	Application Date	Fertilizer application date	Agro	On-Farm Production
28	Fertilizer	Fertilizer Type	Type of fertilizer used	Agro	On-Farm Production
29	Fertilizer	Fertilizer Nature	Fertilizer origin	Agro	On-Farm Production

30	Fertilizer	Nitrogen Analysis %	N content percentage	Agro	On-Farm Production
31	Fertilizer	Ammonia Percentage	Ammonia content	Agro	On-Farm Production
32	Fertilizer	Nitric Percentage	Nitric content	Agro	On-Farm Production
33	Fertilizer	Urea Percentage	Urea content	Agro	On-Farm Production
34	Fertilizer	Phosphate Analysis %	P content percentage	Agro	On-Farm Production
35	Fertilizer	Potassium Analysis %	K content percentage	Agro	On-Farm Production
36	Fertilizer	Manufactured In	Manufacturing location	Agro	On-Farm Production
37	Fertilizer	None CFT If other: Manual Entry Nitrogen%	Manual entry of Nitrogen % if Cool Farm Tool is not automatically collecting the data	Agro	On-Farm Production
38	Fertilizer	None CFT If other: Manual Entry Phosphate%	Manual entry of Phosphate % if Cool Farm Tool is not automatically collecting the data	Agro	On-Farm Production
39	Fertilizer	None CFT If other: Manual Entry Potassium%	Manual entry of Potassium % if Cool Farm Tool is not automatically collecting the data	Agro	On-Farm Production

40	Fertilizer	Application Rate	Rate of application	Agro	On-Farm Production
41	Fertilizer	Application Method	How fertilizer was applied	Agro	On-Farm Production
42	Fertilizer	Emissions Inhibitors	Emissions inhibitor use	Agro	On-Farm Production
43	Fertilizer	N - Application Date	Nitrogen application date	Agro	On-Farm Production
44	Fertilizer	P - Application Date	Phosphorus application date	Agro	On-Farm Production
45	Fertilizer	K - Application Date	Potassium application date	Agro	On-Farm Production
46	Grower Master Data	Bank Details	Bank account information of the grower	Finance	Commercial
47	Planting	Seed Treatments	Seed treatment applied	Agro	On-Farm Production
48	Planting	Treated Against Storage Diseases (Imazalil, Thiabendazool)?	Storage disease treatment status	Agro	On-Farm Production
49	Planting	Treatment done with a fungicide against rhizoctonia?	Rhizoctonia fungicide application	Agro	On-Farm Production

50	Storage Chemicals	Other Application Method - Manual entry	Manual entry for other methods	Agro	On-Farm Production
51	Storage Chemicals	Active Ingredient Application Rate	Application rate of active ingredient	Agro	On-Farm Production

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Foundation Critical / Foundation Recommended / Advanced Insights	Input Format + Validation	Example	Mandatory for KPI calculation?	Yield Optimizer Produce Driver	Satellite Monitoring
Foundation Critical	Dropdown (based on user tables), Not null	Sujit's Farms	#ERROR!	Mandatory	
Foundation Critical	Dropdown (based on user tables), Variety master list, In master list, not null	Low Glyciene	#ERROR!	Mandatory	
Foundation Critical	ISO Country Code, Not null, valid ISO code	IND (India)	#ERROR!		
Foundation Critical	Dropdown (based on user tables), Vendor list, Not null	John Doe	#ERROR!		
Foundation Critical	Dropdown (based on user tables), Not null	John Doe	#ERROR!	Mandatory	
Foundation Critical	Calendar Year (YYYY), Valid year format	Summer	#ERROR!		
Foundation Critical	Polygon Mapping GPS Coordinates (Lat, Long in decimals), Valid GPS format	18:20:23 30:10:12	#ERROR!	Mandatory	Mandatory
Foundation Recommended	Dropdown (based on user tables), Region/State list, Not null	AMESA	#ERROR!	Optional	
Foundation Recommended	Number, Valid contract format	+91-8030273338	#ERROR!		

Foundation Recommended	Dropdown (based on user tables), Chemical list, In chemical list	-	#ERROR!		
Foundation Recommended	Dropdown (based on user tables), Valid input	-	#ERROR!		
Foundation Recommended	Dropdown (Broadcast, Banding, Foliar, Drip), Not null, in allowed values	Broadcast	#ERROR!		
Advanced Insights	Dropdown (based on user tables), Chemical product list, Not null, in product list	-	#ERROR!		
Advanced Insights	Dropdown (Fungicide/Insecticide/etc), Herbicide / Insecticide / Fungicide / Nematicide / Other / Acaricide / Bactericide /	Fungicide	#ERROR!		
Advanced Insights	Dropdown (based on user tables), Producer list, Not null	-	#ERROR!		
Advanced Insights	Dropdown (based on user tables), Ingredient list, Not null	-	#ERROR!		
Advanced Insights	litres/ha Decimal, >=0	25.21	#ERROR!		
Advanced Insights	kg/ha Decimal, >=0	10.45	#ERROR!		
Advanced Insights	Dropdown (based on user tables), Not null	-	#ERROR!		

Advanced Insights	Dropdown (Drip/Flood/Spray/Rainfed), Not null, in allowed values	Drip	#ERROR!	Optional	
Advanced Insights	kg/ha Decimal, >=0, not null	25.21	#ERROR!		
Advanced Insights	Weight in tons ('000kg), >=0, not null	25.21	#ERROR!	Mandatory	
Advanced Insights	Weight in tons ('000kg), >=0, not null	25.21	#ERROR!	Mandatory	
Advanced Insights	Weight in tons ('000kg), >=0, not null	25.21	#ERROR!	Mandatory	
Advanced Insights	yes/no, In allowed values	Yes	#ERROR!		
Advanced Insights	Dropdown (based on user tables), Valid input	-	#ERROR!		
Advanced Insights	Date (YYYY-MM-DD), Valid date, Not null	2025/03/10	#ERROR!	Optional	
Advanced Insights	Dropdown (based on user tables), Specific User-table per Region/Country database and "Other" option available into the drop-down list. Not null in	Product name	#ERROR!	Optional	
Advanced Insights	Dropdown (based on user tables), Not null, in allowed values	Mineral	#ERROR!		

Advanced Insights	Percentage, 0-100	20	#ERROR!		
Advanced Insights	Percentage, 0-100	38.92	#ERROR!		
Advanced Insights	Percentage, 0-100	38.92	#ERROR!		
Advanced Insights	Percentage, 0-100	38.92	#ERROR!		
Advanced Insights	Percentage, 0-100	38.92	#ERROR!		
Advanced Insights	Percentage, 0-100	38.92	#ERROR!		
Advanced Insights	Percentage, 0-100	38.92	#ERROR!		
Advanced Insights	Dropdown (based on user tables), Country/Region list, Valid country/region	-	#ERROR!		
Advanced Insights	Percentage, 0-100	38.92	#ERROR!		
Advanced Insights	Percentage, 0-100	38.92	#ERROR!		
Advanced Insights	Percentage, 0-100	38.92	#ERROR!		

Advanced Insights	kg/ha Decimal, >=0, not null	25.21	#ERROR!		
Advanced Insights	Dropdown (Broadcast, Banding, Foliar, Drip) Dropdown (Broadcast, Incorporate in-furrow, Fertigation, Sprayer). Not null	-	#ERROR!		
Advanced Insights	yes/no, In allowed values	Yes	#ERROR!		
Advanced Insights	Date (YYYY-MM-DD), Valid date, Not null	2025/03/10	#ERROR!	Optional	
Advanced Insights	Date (YYYY-MM-DD), Valid date, Not null	2025/03/10	#ERROR!	Optional	
Advanced Insights	Date (YYYY-MM-DD), Valid date, Not null	2025/03/10	#ERROR!	Optional	
Advanced Insights	Alpha-Numeric, n/a, Valid format	RFX1001011010	#ERROR!		
Advanced Insights	Dropdown (based on user tables), Treatment types list, In treatment list	-	#ERROR!		
Advanced Insights	yes/no, In allowed values	Yes	#ERROR!		
Advanced Insights	yes/no, In allowed values	Yes	#ERROR!		

Advanced Insights	Dropdown (based on user tables), Valid text	-	#ERROR!		
Advanced Insights	kg/ha or litres/ha Decimal, >=0	25.21	#ERROR!		

Source System (SAP)	Source Table (SAP)	Benchmarks	Actions based on benchmarks
Source System	Source Table		
Variety System	Variety Master		
Master Data	Country Master		
Master Data	Vendor Master		
Farm System	Grower Master		
Farm System	Season Master		
Farm System	Field Master		
Master Data	Region Master		
Master Data	Vendor Master		

Storage System	Chemical Log		
Storage System	Chemical Log		
Farm System	Fertilizer Logs		
Farm System	Chemical Logs		
Farm System	Chemical Logs		
Farm System	Chemical Logs		
Farm System	Chemical Logs		
Farm System	Chemical Logs		
Farm System	Chemical Logs		

Farm System	Field Season		
Farm System	Fertilizer Logs		
Farm System	Fertilizer Logs		
Farm System	Fertilizer Logs		
Farm System	Fertilizer Logs		
Farm System	Field Season		
Farm System	Fertilizer Logs		
Farm System	Fertilizer Logs		
Farm System	Fertilizer Logs		

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Farm System	Fertilizer Logs		
Farm System	Fertilizer Logs		
Farm System	Grower Master		
Farm System	Seed Data		
Farm System	Seed Data		
Farm System	Seed Data		

Storage System	Chemical Log		
Storage System	Chemical Log		